



CSCI 2800

RPISEC

Fall 2026

```
; ***** SUBROUTINE *****  
  
EXPORT __libc_csu_init  
__libc_csu_init                ; DATA XREF: _start+1Ato  
                                ; .text:initto  
  
    PUSH.W                     {R3-R9,LR}  
    MOV                         R7, R0  
    LDR                         R6, =( __do_global_dtors_aux_fini  
    MOV                         R8, R1  
    LDR                         R5, =( __frame_dummy_init_array_ent  
    MOV                         R9, R2  
    ADD                         R6, PC ; __do_global_dtors_aux_fi  
    BLX                         .init_proc  
    ADD                         R5, PC ; __frame_dummy_init_array  
    SUBS                        R6, R6, R5  
    ASRS                        R6, R6, #2  
    BEQ                         locret_104F8  
    MOVS                        R4, #0  
  
loc_104E6                      ; CODE XREF: __libc_csu_in  
    ADDS                        R4, #1  
    LDR.W                       R3, [R5],#4  
    MOV                         R2, R9  
    MOV                         R1, R8  
    MOV                         R0, R7  
    BLX                         R3 ; frame_dummy  
    CMP                         R4, R6  
    BNE                         loc_104E6  
  
locret_104F8                   ; CODE XREF: __libc_csu_in  
    POP.W                       {R3-R9,PC}  
; End of function __libc_csu_init  
  
; -----  
off_104FC                      DCD __do_global_dtors_aux_fini_array_entry - 0x104  
                                ; DATA XREF: __libc_csu_in  
off_10500                      DCD __frame_dummy_init_array_entry - 0x104E0  
                                ; DATA XREF: __libc_csu_in  
  
; ***** SUBROUTINE *****  
  
EXPORT __libc_csu_fini  
__libc_csu_fini                ; DATA XREF: _start+10to  
                                ; .text:off_10398to  
  
    BX                         LR  
; End of function __libc_csu_fini  
  
; -----
```

RPISEC

- The Computer Security Club @ RPI
- We **compete** in CTFs
- We **teach** security (hacking)
 - Weekly Wednesday & Friday Meetings
 - Wednesday 6PM in Lally 102: Introductory
 - Friday 6PM in LOW 4050: Advanced
 - CTFs, Guest Speakers, Special Events

```
; ***** SUBROUTINE *****  
  
EXPORT __libc_csu_init  
__libc_csu_init  
    ; DATA XREF: _start+1Ato  
    ; .text:00000000  
    PUSH.W    {R3-R9,LR}  
    MOV       R7, R0  
    LDR       R6, =( __do_global_dtors_aux_fini_array_entry  
    MOV       R8, R1  
    LDR       R5, =( frame_dummy_init_array_entry  
    MOV       R9, R2  
    ADD       R6, PC ; __do_global_dtors_aux_fini_array_entry  
    BLX       .init_proc  
    ADD       R5, PC ; __frame_dummy_init_array_entry  
    SUBS      R6, R5  
    ASRS      R6, R6, #2  
    BEQ       locret_104F8  
    MOVS      R4, #0  
  
loc_104E6  
    ; CODE XREF: __libc_csu_init+00000000  
    ADDS      R4, #1  
    LDR.W     R3, [R5],#4  
    MOV       R2, R9  
    MOV       R1, R8  
    MOV       R0, R7  
    BLX       R3 ; frame_dummy  
    CMP       R4, R6  
    BNE       loc_104E6  
  
loc_104F8  
    ; CODE XREF: __libc_csu_init+00000004  
    POP.W     {R3-R9,PC}  
; End of function __libc_csu_init  
  
;-----  
off_104FC DCD __do_global_dtors_aux_fini_array_entry - 0x104FC  
; DATA XREF: __libc_csu_init+0000000C  
off_10500 DCD __frame_dummy_init_array_entry - 0x10500  
; DATA XREF: __libc_csu_init+00000010  
; ***** SUBROUTINE *****  
  
EXPORT __libc_csu_fini  
__libc_csu_fini  
    ; DATA XREF: _start+10to  
    ; .text:00000010  
    BX        LR  
; End of function __libc_csu_fini  
;-----
```

Security Topics

- Reverse Engineering
- Binary Exploitation
- Web Exploitation
- Cryptography
- Forensics
- Malware Analysis & Development
- Embedded Systems
- Kernel Exploitation
- Hardware RE

Fall
Spring

and
more...

Industry

- Microsoft
- Google
- Meta (Facebook)
- Apple
- Oracle
- Bloomberg
- Tumblr
- NCC Group
- Trail of Bits
- Palo Alto Networks
- Mandiant/FireEye
- Siege Technologies
- Exodus Intelligence
- White House

- MIT Lincoln Lab
- MITRE
- Lockheed Martin
- Raytheon
- General Dynamics
- NSA
- CISCO

All places we have alumni,
internships, and connections

Why RPISEC?

- Motivated Students
 - Student run, less formal
- Alumni
 - Dedicated to club
 - Funding
 - Trainings
- Internships
- CTFs
 - Quantity, Quality, Difficulty
 - Hardware!
- Travel

```
; ***** SUBROUTINE *****  
  
EXPORT __libc_csu_init  
__libc_csu_init  
    ; DATA XREF: __start+1Ato  
    ; .text:00000000  
    PUSH.W    {R3-R9,LR}  
    MOV       R7, R0  
    LDR       R6, =(__do_global_dtors_aux_fini_array_entry - 0x104FC)  
    MOV       R8, R1  
    LDR       R5, =(__frame_dummy_init_array_entry - 0x104E0)  
  
    ; End of function __libc_csu_init  
    BX        LR  
; *****  
; End of function __libc_csu_fini  
; *****
```

🔍 Internship Acquired? ▼

Yes	▼
Yes	▼
Yes	▼
Yes	▼
Yes	▼
Yes	▼
Yes	▼
Yes	▼
Yes	▼
Yes	▼
Yes	▼
Yes	▼
NOT YET	▼
NOT YET	▼
NOT YET	▼

Shmooscon, Washington, D.C.

```
EXPORT __libc_csu_init
; DATA XREF: _start+1Ato
; .text:00000000
libcsu_init
; {R3-R9,LR}
PUSH {R7, R0}
LDR R6, =(__do_global_dtors_aux_fini_array_entry)
MOV R8, R1
LDR R5, =(__frame_dummy_init_array_entry)
MOV R9, R2
ADD R6, PC ; __do_global_dtors_aux_fini_array_entry
BLX .init_proc
ADD R5, PC ; __frame_dummy_init_array_entry
SUBS R6, R6, R5
ASRS R6, R6, #2
```



```
DCD __do_global_dtors_aux_fini_array_entry - 0x104E0
; DATA XREF: __libc_csu_init
DCD __frame_dummy_init_array_entry - 0x104E0
; DATA XREF: __libc_csu_init
```

; ***** SUBROUTINE *****

```
EXPORT __libc_csu_fini
__libc_csu_fini
; DATA XREF: _start+10to
; .text:00000000
```

```
BX LR
; End of function __libc_csu_fini
```


BsidesROC, Rochester NY



Upcoming Events!

- **Weekly** Wednesday/Friday Meetings
 - Wednesday is beginner
 - Friday is advanced (everyone is welcome!)
- September 25th, Alexei Bulazel
 - Special Assistant to the President
 - NSC Director for Cyber

Semester
Calendar Coming
Maybe!

Contact

Discord - <https://discord.gg/FSssMc3NYt>

Slack - rpiSEC.slack.com

Website - rpiSEC.ec



A Quick Example - gets()

- Similar to scanf()
- HIGHLY UNSAFE
- Allows us to write past the end of a buffer

```
; ***** SUBROUTINE *****
libc_csu_init
EXPORT __libc_csu_init
; DATA XREF: _start+1Ato
; .text:00000000
PUSH.W      {R3-R9,LR}
MOV         R7, R0
LDR         R6, =(__do_global_dtors_aux_fini_array_entry)
MOV         R8, R1
LDR         R5, =(__frame_dummy_init_array_entry)
MOV         R9, R2
ADD         R6, PC ; __do_global_dtors_aux_fini_array_entry
BLX         .init_proc
ADD         R5, PC ; __frame_dummy_init_array_entry
SUBS        R6, R6, R5
ASRS        R6, R6, #2
BEQ         locret_104F8
MOVS        R4, #0

loc_104E6
; CODE XREF: __libc_csu_init
ADDS        R4, #1
LDR.W       R3, [R5], #4
MOV         R2, R9
MOV         R1, R8
MOV         R0, R7
CMP         R3, R4 ; frame_dummy
BNE         loc_104E6

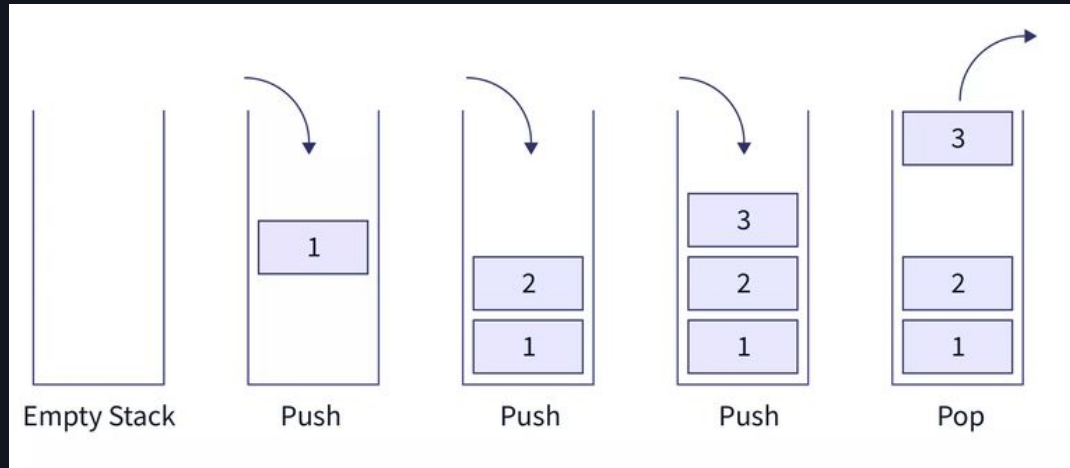
locret_104F8
; CODE XREF: __libc_csu_init
POP.W       {R3-R9,PC}
; End of function __libc_csu_init

; *****
off_104FC    DCD __do_global_dtors_aux_fini_array_entry - 0x104FC
; DATA XREF: __libc_csu_init
off_10500    DCD __frame_dummy_init_array_entry - 0x10500
; DATA XREF: __libc_csu_init

; ***** SUBROUTINE *****
__libc_csu_fini
EXPORT __libc_csu_fini
; DATA XREF: _start+10to
; .text:00000000
BX          LR
; End of function __libc_csu_fini
; *****
```

Stack Review

- Variables declared first are on the bottom



fairgame.rpis.ec

- Sign up, try some CTF challenges.
 - Use your RPI Email!

THX FOR COMING!